SAVANNA SUDAN CLIMATE

SAVANNA TRANSITION MAP → ITCZ SEASONAL SWITCH → WATER-RELIABILITY CHAIN → GRASS-TREE GRADIENT → FIRE-HERBIVORY BALANCE → WILDLIFE-LIVELIHOOD MATRIX

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 SAVANNA TRANSITION MAP

SAVANNA TRANSITION MAP | WETTER AND TALLER GRASS | SUDAN BELT | CLASSIC WET-DRY SAVANNA | SHORTER GRASS AND OPEN TREES | DESERT MARGIN | DROUGHT RISK INTENSIFIES | MUST REMEMBER

EQUATORWARD → wetter and taller grass → SUDAN BELT → classic wet-dry savanna → POLEWARD → shorter grass and open trees → DESERT MARGIN → drought risk intensifies → MUST REMEMBER: Savanna Aw/As climates reflect seasonal ITCZ migration between convective wet summers and subtropical-high

CORE CLAIM

- EQUATORWARD → wetter and taller grass
- SUDAN BELT → classic wet-dry savanna

MECHANISM

- POLEWARD → shorter grass and open trees
- DESERT MARGIN → drought risk intensifies

CONSEQUENCE / LIMIT

- MUST REMEMBER: Savanna Aw/As climates reflect seasonal ITCZ migration between convective wet summers and subtropical-high dry
- winters, sustaining tall grasses with scattered drought/fire-adapted trees across Africa, South America and northern Australia.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Savanna Aw/As climates reflect seasonal ITCZ migration between convective wet summers and subtropical-high dry winters, sustaining tall grasses with scattered drought/fire-adapted trees across Africa, South America and northern Australia.

01

STAGE 01 ITCZ SEASONAL SWITCH

ITCZ SEASONAL SWITCH | HIGH-SUN SEASON | ITCZ APPROACHES | CONVERGENCE + CONVECTION | SUMMER RAIN | LOW-SUN SEASON | DRY TRADES DOMINATE | HOT WET

HIGH-SUN SEASON → ITCZ approaches

CONVERGENCE + CONVECTION → summer rain → LOW-SUN SEASON → dry trades dominate → RESULT → hot wet / warm dry rhythm

CORE CLAIM

- HIGH-SUN SEASON → ITCZ approaches
- CONVERGENCE + CONVECTION → summer rain

MECHANISM

- LOW-SUN SEASON → dry trades dominate
- RESULT → hot wet / warm dry rhythm

CONSEQUENCE / LIMIT

- HIGH-SUN SEASON → ITCZ approaches
- CONVERGENCE + CONVECTION → summer rain

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESULT → hot wet / warm dry rhythm.

02

STAGE 02 WATER-RELIABILITY CHAIN

WATER-RELIABILITY CHAIN | LATE ONSET | DELAYED SOWING | MID-SEASON BREAK | CROP AND FORAGE STRESS | EARLY CESSATION | INCOMPLETE GRAIN FILLING | MIGRATION AND HERD LOSS RISK

LATE ONSET → delayed sowing → MID-SEASON BREAK → crop and forage stress → EARLY CESSATION → incomplete grain filling → DROUGHT → migration and herd loss risk

CORE CLAIM

- LATE ONSET → delayed sowing
- MID-SEASON BREAK → crop and forage stress

MECHANISM

- EARLY CESSATION → incomplete grain filling
- DROUGHT → migration and herd loss risk

CONSEQUENCE / LIMIT

- LATE ONSET → delayed sowing
- MID-SEASON BREAK → crop and forage stress

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EARLY CESSATION → incomplete grain filling; DROUGHT → migration and herd loss risk.

03

STAGE 03 GRASS-TREE GRADIENT

GRASS-TREE GRADIENT | WET MARGIN | TALL GRASS + MORE TREES | CORE SAVANNA | PARKLAND MOSAIC | DRY MARGIN | SHORT GRASS + THORNY SHRUBS | DESERT EDGE

WET MARGIN → tall grass + more trees

CORE SAVANNA → parkland mosaic → DRY MARGIN → short grass + thorny shrubs → DESERT EDGE → discontinuous vegetation

- LATE ONSET → delayed sowing
- MID-SEASON BREAK → crop and forage stress

- EARLY CESSATION → incomplete grain filling
- DROUGHT → migration and herd loss risk

- LATE ONSET → delayed sowing
- MID-SEASON BREAK → crop and forage stress

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EARLY CESSATION → incomplete grain filling; DROUGHT → migration and herd loss risk.

03

STAGE 03 GRASS-TREE GRADIENT

GRASS-TREE GRADIENT | WET MARGIN | TALL GRASS + MORE TREES | CORE SAVANNA | PARKLAND MOSAIC | DRY MARGIN | SHORT GRASS + THORNY SHRUBS | DESERT EDGE

WET MARGIN → tall grass + more trees

CORE SAVANNA → parkland mosaic → DRY MARGIN → short grass + thorny shrubs → DESERT EDGE → discontinuous vegetation

CORE CLAIM
- WET MARGIN → tall grass + more trees
- CORE SAVANNA → parkland mosaic

MECHANISM
- DRY MARGIN → short grass + thorny shrubs
- DESERT EDGE → discontinuous vegetation

CONSEQUENCE / LIMIT
- WET MARGIN → tall grass + more trees
- CORE SAVANNA → parkland mosaic

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DRY MARGIN → short grass + thorny shrubs.

04

STAGE 04 FIRE-HERBIVORY BALANCE

FIRE-HERBIVORY BALANCE | SEASONAL RAIN | GRASS BIOMASS | DRY SEASON | COMBUSTIBLE FUEL | FIRE + BROWSING | SEEDLING SUPPRESSION | THICK BARK + ROOTS

SEASONAL RAIN → grass biomass → DRY SEASON → combustible fuel → FIRE + BROWSING → seedling suppression → THICK BARK + ROOTS → selected tree survival

CORE CLAIM
- SEASONAL RAIN → grass biomass
- DRY SEASON → combustible fuel

MECHANISM
- FIRE + BROWSING → seedling suppression
- THICK BARK + ROOTS → selected tree survival

CONSEQUENCE / LIMIT
- SEASONAL RAIN → grass biomass
- DRY SEASON → combustible fuel

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: THICK BARK + ROOTS → selected tree survival.

05

STAGE 05 WILDLIFE-LIVELIHOOD MATRIX

WILDLIFE-LIVELIHOOD MATRIX | FOLLOW SEASONAL FORAGE AND WATER | TRACK HERBIVORE CONCENTRATIONS | MOBILITY SPREADS RISK | SORGHUM TIMING

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
HERDS → follow seasonal forage and water	PASTORALISTS → mobility spreads risk	HERDS → follow seasonal forage and water
PREDATORS → track herbivore concentrations	CULTIVATORS → millet / sorghum timing	PREDATORS → track herbivore concentrations

CORE CLAIM
- HERDS → follow seasonal forage and water
- PREDATORS → track herbivore concentrations

MECHANISM
- PASTORALISTS → mobility spreads risk
- CULTIVATORS → millet / sorghum timing

CONSEQUENCE / LIMIT
- HERDS → follow seasonal forage and water
- PREDATORS → track herbivore concentrations

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PASTORALISTS → mobility spreads risk; CULTIVATORS → millet / sorghum timing.

06

STAGE 06 DEGRADATION PATHWAY

DEGRADATION PATHWAY | COVER LOSS | RAINDROP AND WIND EXPOSURE | WET SEASON | RUNOFF, LEACHING AND EROSION | DRY SEASON | FIRE AND SURFACE SEALING | FIXED WATER POINT

COVER LOSS → raindrop and wind exposure → WET SEASON → runoff, leaching and erosion → DRY SEASON → fire and surface sealing → FIXED WATER POINT → local grazing concentration

CORE CLAIM
- COVER LOSS → raindrop and wind exposure
- WET SEASON → runoff, leaching and erosion

MECHANISM
- DRY SEASON → fire and surface sealing
- FIXED WATER POINT → local grazing concentration

CONSEQUENCE / LIMIT
- COVER LOSS → raindrop and wind exposure
- WET SEASON → runoff, leaching and erosion

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DRY SEASON → fire and surface sealing; FIXED WATER POINT → local grazing concentration.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PASTORALISTS → mobility spreads risk; CULTIVATORS → millet / sorghum timing.

06

STAGE 06

DEGRADATION PATHWAY

DEGRADATION PATHWAY | COVER LOSS | RAINDROP AND WIND EXPOSURE | WET SEASON | RUNOFF, LEACHING AND EROSION | DRY SEASON | FIRE AND SURFACE SEALING | FIXED WATER POINT

COVER LOSS → raindrop and wind exposure → WET SEASON → runoff, leaching and erosion → DRY SEASON → fire and surface sealing → FIXED WATER POINT → local grazing concentration

CORE CLAIM
- COVER LOSS → raindrop and wind exposure
- WET SEASON → runoff, leaching and erosion

MECHANISM
- DRY SEASON → fire and surface sealing
- FIXED WATER POINT → local grazing concentration

CONSEQUENCE / LIMIT
- COVER LOSS → raindrop and wind exposure
- WET SEASON → runoff, leaching and erosion

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DRY SEASON → fire and surface sealing; FIXED WATER POINT → local grazing concentration.

07

STAGE 07

INDIA FOREST GRADIENT

INDIA FOREST GRADIENT | MOIST DECIDUOUS | LONGER WET SEASON | DRY DECIDUOUS | STRONGER LEAF-SHEDDING | SEMI-ARID MOISTURE STRESS | OVERLAPPING BELTS, NOT FIXED WALLS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
MOIST DECIDUOUS → longer wet season	THORN / SCRUB → semi-arid moisture stress	MOIST DECIDUOUS → longer wet season
DRY DECIDUOUS → stronger leaf-shedding	RULE → overlapping belts, not fixed walls	DRY DECIDUOUS → stronger leaf-shedding

CORE CLAIM
- MOIST DECIDUOUS → longer wet season
- DRY DECIDUOUS → stronger leaf-shedding

MECHANISM
- THORN / SCRUB → semi-arid moisture stress
- RULE → overlapping belts, not fixed walls

CONSEQUENCE / LIMIT
- MOIST DECIDUOUS → longer wet season
- DRY DECIDUOUS → stronger leaf-shedding

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: THORN / SCRUB → semi-arid moisture stress; RULE → overlapping belts, not fixed walls.

08

STAGE 08

INDIAN GRASSLAND FIREWALL

INDIAN GRASSLAND FIREWALL | HUMID TALL GRASS AND FLOODPLAIN | MONTANE GRASS-FOREST MOSAIC | ARID-SALINE KACHCHH GRASSLAND | HYDROLOGY AND HISTORY FIRST

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
TERAI → humid tall grass and floodplain	BANNI → arid-saline Kachchh grassland	TERAI → humid tall grass and floodplain
SHOLA → montane grass-forest mosaic	MANAGEMENT → hydrology and history first	SHOLA → montane grass-forest mosaic

CORE CLAIM
- TERAI → humid tall grass and floodplain
- SHOLA → montane grass-forest mosaic

MECHANISM
- BANNI → arid-saline Kachchh grassland
- MANAGEMENT → hydrology and history first

CONSEQUENCE / LIMIT
- TERAI → humid tall grass and floodplain
- SHOLA → montane grass-forest mosaic

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BANNI → arid-saline Kachchh grassland; MANAGEMENT → hydrology and history first.

09

STAGE 09

RESTORATION DECISION TREE

RESTORATION DECISION TREE | NATURAL OPEN ECOSYSTEM | CONSERVE OPENNESS | DEGRADED SCRUB | RESTORE NATIVE MOSAIC | WATER PROCESS | RECHARGE AND DRAINAGE | GRAZING AND TENURE DESIGN

NATURAL OPEN ECOSYSTEM → conserve openness

DEGRADED SCRUB → restore native mosaic → WATER PROCESS → recharge and drainage → LIVELIHOOD → grazing and tenure design

CORE CLAIM
- NATURAL OPEN ECOSYSTEM → conserve openness
- DEGRADED SCRUB → restore native mosaic

MECHANISM
- WATER PROCESS → recharge and drainage
- LIVELIHOOD → grazing and tenure design

CONSEQUENCE / LIMIT
- CLOSE DISTINCTION: Savanna is tropical grassland, steppe is semi-arid temperate/subtropical grassland, and parkland structure is not
- evidence of a climax maintained by climate alone because fire, herbivory and land use matter.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Savanna is tropical grassland, steppe is semi-arid temperate/subtropical grassland, and parkland structure is not evidence of a climax maintained by climate alone because fire, herbivory and land use matter.

RESTORATION DECISION TREE | NATURAL OPEN ECOSYSTEM | CONSERVE OPENNESS | DEGRADED SCRUB | RESTORE NATIVE MOSAIC | WATER PROCESS | RECHARGE AND DRAINAGE | GRAZING AND TENURE DESIGN

NATURAL OPEN ECOSYSTEM → conserve openness

DEGRADED SCRUB → restore native mosaic → WATER PROCESS → recharge and drainage → LIVELIHOOD → grazing and tenure design

CORE CLAIM
- NATURAL OPEN ECOSYSTEM → conserve openness
- DEGRADED SCRUB → restore native mosaic

MECHANISM
- WATER PROCESS → recharge and drainage
- LIVELIHOOD → grazing and tenure design

CONSEQUENCE / LIMIT
- CLOSE DISTINCTION: Savanna is tropical grassland, steppe is semi-arid temperate/subtropical grassland, and parkland structure is not
- evidence of a climax maintained by climate alone because fire, herbivory and land use matter.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Savanna is tropical grassland, steppe is semi-arid temperate/subtropical grassland, and parkland structure is not evidence of a climax maintained by climate alone because fire, herbivory and land use matter.

10

STAGE 10 SAVANNA PYQ FIREWALL

SAVANNA PYQ FIREWALL | 2021 PRELIMS | TREE-LIMITING CONDITIONS | LONG DRY SEASON | FIRE + HERBIVORY | NO ANSWER LETTER INVENTED | EVIDENCE LIMIT | FIRE CAN RECYCLE NUTRIENTS AND MAINTAIN OPENNESS BUT FREQUENCY/INTENSITY

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
2021 PRELIMS → tree-limiting conditions	DISTURBANCE → fire + herbivory	EVIDENCE LIMIT: Fire can recycle nutrients and maintain openness but frequency/intensity changes may degrade systems
CLIMATE → long dry season	LEDGER → no answer letter invented	Indian deciduous forests and grasslands need region-specific rainfall, fire and grazing evidence rather than direct Sudan-type equivalence.

CORE CLAIM
- 2021 PRELIMS → tree-limiting conditions
- CLIMATE → long dry season

MECHANISM
- DISTURBANCE → fire + herbivory
- LEDGER → no answer letter invented

CONSEQUENCE / LIMIT
- EVIDENCE LIMIT: Fire can recycle nutrients and maintain openness but frequency/intensity changes may degrade systems
- Indian deciduous forests and grasslands need region-specific rainfall, fire and grazing evidence rather than direct Sudan-type equivalence.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: Fire can recycle nutrients and maintain openness but frequency/intensity changes may degrade systems; Indian deciduous forests and grasslands need region-specific rainfall, fire and grazing evidence rather than direct Sudan-type equivalence.

11

STAGE 11 SAVANNA ANSWER SPINE

SAVANNA ANSWER SPINE | RAINFOREST-DESERT TRANSITION | ITCZ AND WET-DRY SEASON | GRASS-TREE DISTURBANCE BALANCE | INDIA MOSAICS AND RESTORATION

LOCATE → rainforest-desert transition → DERIVE → ITCZ and wet-dry season → EXPLAIN → grass-tree disturbance balance → APPLY → India mosaics and restoration

CORE CLAIM
- LOCATE → rainforest-desert transition
- DERIVE → ITCZ and wet-dry season

MECHANISM
- EXPLAIN → grass-tree disturbance balance
- APPLY → India mosaics and restoration

CONSEQUENCE / LIMIT
- LOCATE → rainforest-desert transition
- DERIVE → ITCZ and wet-dry season

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EXPLAIN → grass-tree disturbance balance; APPLY → India mosaics and restoration.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | INDIA'S SAVANNA ANALOGUE | DRY DECIDUOUS FOREST + THORN SCRUB + GRASSLANDS . | TROPICAL MOIST DECIDUOUS FORESTS | 100–200 CM RAINFALL; DOMINANT TEAK AND SAL . | MOIST DECIDUOUS REGIONS | SAHYADRIS , NE PENINSULA, HIMALAYAN FOOTHILLS. | THORN/SCRUB REFLECTS ARIDITY PLUS LAND-USE PRESSURE; IT IS NOT

OPTIONAL SOURCE DEPTH
- India's savanna analogue = dry deciduous forest + thorn scrub + grasslands .
- Tropical moist deciduous forests: 100–200 cm rainfall; dominant teak and sal .
- Moist deciduous regions: Sahyadris , NE peninsula, Himalayan foothills.
- Thorn/scrub reflects aridity plus land-use pressure; it is not invariably degraded moist deciduous forest.

ADVANCED DEBATE
- Rainfall thresholds overlap regionally; use approximate belts rather than rigid universal cut-offs.
- Thorn forest regions: peninsular India, Rajasthan, Haryana, Punjab, W UP, Kachchh, MP, Himalayan foothills.
- Deciduous trees shed leaves in dry season.
- Grasslands include Terai, shola and Banni examples in the source notes.

LEGACY / NUANCE
- Fire and grazing maintain open grassland/woodland mosaics.
- NOT: Dry deciduous forests are evergreen → They are deciduous and shed leaves in the dry season.
- NOT: Thorn forests occur in the wettest parts of India → They occur in arid/semi-arid belts, generally
- NOT: Banni and Terai grasslands are the same type → Terai is humid tall wet grassland; Banni is arid/saline Kachchh grassland.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels